

A fuzzy recommendation system for predicting the customers interests using sentiment analysis and ontology in e-commerce

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ABSTRACT

In Electronic commerce, customer reviews play a significant role in purchase making decision. Most of the existing recommendation systems consider the customer reviews, user purchase history and product rating for predicting the recommended product. Since the users interest are varying over time, the existing recommendation systems lack in finding the current relevant items to the customers. To overcome this problem, this article proposes a new fuzzy logic-based product recommendation system which dynamically predicts the most relevant products to the customers in online shopping according to the users' current interests. A novel algorithm has been proposed in this paper for computing the sentimental score of the product with associated end user target category. Finally, the proposed fuzzy rules and ontology-based recommendation system uses ontology alignment for making decisions that are more accurate and predict dynamically based on the search context. The experimental results of the proposed recommendation system show better performance than the existing product recommendation systems in terms of prediction accuracy of the relevant products for target users and in the time taken to provide such recommendations.

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1. Introduction

In the recent years, due to high internet penetration and rapid growth in mobile usage, online shopping has increased tremendously and has become very popular [1]. Customers prefer online shopping as they can take their purchasing decisions on the products by referring to the review comments, technical specification and product rating.

Customers are also attracted by cash back offers, cash on delivery and easy replacement guarantees. Moreover, the time constraint in shopping at store and road traffic are key factors to prefer online shopping in major cities as the customers can quickly place order in online and get the products delivered easily at door step. The popularity of online shopping also gave business opportunities to a lot of small-scale suppliers to sell their products without any major investment for infrastructure and promotion.

On the other hand, the organizations are incorporating Business Intelligence (BI) in the e-commerce domain to understand more about their end users interests. This information helps the organization to improve customer satisfaction and increases their sales and profit. Now, the e-commerce is extended till agriculture

domain [2] by setting up the business platform and trading the cultivated products online. This platform plays a significant role in the reduction of poverty of poor farmers especially in the rural areas. E-commerce market is also contributing to the online virtual games which suits for the user age group and DevOps platform [3].

In an e-commerce domain, the recommendation system helps both the buyer and the seller community. Recommendation systems are necessary to get relevant products and more personalized recommendation based on the user purchase behavior and interest from huge number of products [4]. Collaborative and content-based recommendation methods are widely used recommendation techniques. Recently, the demographic recommendation and context-based recommendation techniques give more relevant and more interesting recommended products to the customer.

E-learning education recommendation system are developed [5] to recommend the e-learning courses based on individual learning interest. Customer reviews are the customer opinions and they can be positive or negative. The identification of opinion from the data is called as opinion mining or sentiment analysis. The sentiment analysis [6] is generally performed at three levels namely document level, sentence level and the aspect level. Sentiment classification techniques are broadly classified into lexicon-based approach and machine learning approach. Deep learning techniques [7] are used to identify the sentiments that

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are expressed in the unstructured data. Recently, the sentiment analysis is also performed in multi modal data like micro videos and hash tags [8].

In online shopping product recommendation system, Fuzzy logic helps to improve the prediction accuracy of the recommendation list and gives highly personalized and interested products to the customer [8]. Recently, fuzzy based recommendation systems are applied in government sectors [9] for applications like tax payment, which gives some confidence to increase the awareness of the tax payment among the public. Fuzzy method is used to improve the decision accuracy in decision support system and hence most applications use fuzzy logic for taking decisions to select the best vendor from a set of vendors [10].

Ontological database is primarily preferred to handle large volume of unstructured data and incomplete data. Ontological database store and retrieve incomplete information, also known as "Open World Assumption" (OWA). The main advantage of storing the knowledge in ontological database is to directly apply decision-making process through ontology alignment.

The major contributions of this paper are: (1) It identifies the real target end user group from customer reviews (2) It computes the rating score for different user category for each product (3) It recommends relevant products using Fuzzy recommendation logic and ontology matching through fuzzy rules. (4) A new pre-processing algorithm is introduced for identifying the contributed and meaningful information from the Amazon review dataset which is available as a standard benchmark dataset. (5) An ontological database is created to store the domain knowledge and to perform decision making through reasoning with ontology alignment.

The remainder of this research article is formulated as below: The related works in the direction of e-commerce, sentimental analysis and recommendation system are described in Section 2. Motivation to develop new recommendation system is described in Section 3. The recommendation system framework is sketched neatly with necessary explanation in Section 4. Section 5 contains a brief description about the newly proposed recommendation system. Section 6 demonstrates the experimental results of the proposed recommendation system. Section 7 gives conclusions on the proposed work with future directions for enhancing the proposed work.

2. Literature survey

E-commerce is applied in many areas including business applications, industrial systems and e-learning systems where it is used for setting a new virtual business operation models like new education system with virtual laboratory and classrooms. Vejacka [11] elaborated the opportunity of setting up lot of business environments like virtual bank and e-shops in e-commerce. It enables the businesspersons to do business without major capitalization and infrastructure. In addition to commercial or retail industry, ecommerce is also used in the environment preservation applications as well. Elpawati and Yuni Sugiarti [12] proposed an information system developed along with the e-commerce system to effectively manage waste by using micro organic decomposer.

Tania Ferreira et al. [13] elaborated the influence of using Business intelligence in the e-commerce sector to understand the real customer interest. Application of e-commerce extends the boundary further to online gaming platform. Randy Rachmadi et al. [14] proposed an online game market place which provides the opportunity for virtual games to meet all the user category groups. Virtual education industry is rapidly growing, especially virtual classrooms and online courses play an important role during the pandemic situations. Ni wang et al. [15] designed

a new framework of cross border e-commerce for the online courses, which are application oriented. This enables to have a new dimension of online education model in e-commerce.

Recommendation systems are used to provide relevant products to the customer from a wide range of available products according to user preference. Recommendation techniques are broadly classified into collaborative filtering and content-based techniques. Content-based technique predicts the rating based on previous purchase history. Whereas collaborative filtering does not require any previous knowledge on about users or items, it rather works with interaction between the user and items. Nayana Vaidya et al. [16] described the essentials of recommendation system in e-commerce. Recommendation system not only resolves the information overload problem, but also helps the online users to get personalized recommendations. Moreover, the recommendation systems help online users to make right purchase decisions and also improves the overall shopping experience.

Instead of computing the product ranking by referring individually, Saratchandra et al. [17] elaborated the influence of considering the neighborhood products, above and below available products when computing the product ranking. By considering this delta feature approach, the predication of recommendations is improved when compared with the existing baseline systems and online users are can easily share their shopping and product experience as review comments.

Guoshuai Zhao et al. [18] explored the internal influence of the customer reviews for the social recommendation. Customer ratings and the number of reviews are the external influence factors in predicting the relevant products to the customer. Their results have shown that the user internal influence factor has increased the prediction efficiency of the recommendation system. Gautam Chauhan et al. [19] proposed the Customer-Aware Recommender System in e-commerce, which pushes the personalized notifications to the online users. Their recommendation system uses collaborating filtering with user profile analysis for performing the predictive analytics. By understanding the user behavior and profile information, the predicted relevant recommendations are pushed to the users as notification. This recommendation out performs the similar systems and takes user attention more when compared to usual event or time-based notifications.

Tin-Chih et al. [20] applied the recommendation system in the health care sector for recommending the suitable clinic by grouping the patients based on the influence factors. This approach splits the patients into multiple groups by analyzing the patient's unknown preferences. Opinions expressed in the review plays an important role in taking some decisions to purchase. There are many techniques, which are available to identify the opinions expressed in the tweets. Tweet sentiment analysis helps to rank the products more efficiently. Huyen Trang Phan et al. [21] enhanced the sentiment analysis approach with fuzzy ensemble method by considering the position of polarity words in order to understand the context and phrases of the words. The use of fuzzy sentiments improves the decision performance and it also in turn increases the efficiency of prediction.

Yang et al. [22] proposed a sentiment enhanced multi modal attentive recommendation model for providing predictive recommendations in micro videos. The users are now using pictures and micro videos to express their opinions. It improves the accuracy of the recommendations by identifying and analyzing the sentiment hash tags. The User Item Community Detection based Recommendation (UICDR) method groups the users and items into clusters by constructing user item bipartite network and then Collaborative Filtering method is applied on each cluster to predict the most recommended product [23].

Both content based and collaborative filtering approaches have their own advantages and drawbacks and hence Liang Feng et al.

[24] built a hybrid approach and it provides the advantages of both these approaches. Hongtao Liu et al. [25] developed another Hybrid recommendation model (HRDR) and this model provides the recommendations by referring the information from both ratings and explicit feedback obtained through user reviews by using deep learning approach. The rating matrix suffers when the number of users and products is huge [25]. This model preferred to use implicit feedbacks instead of referring to explicit feedback ratings. In another recommendation system, neural network is used to predict the personalized products with implicit user feedback information [26], which is developed using a multi criteria prediction model. First, this system identifies the criteria rating by deep learning network and then predicts the recommendations with the rating.

To improve the accuracy of predictions, this recommendation system used the contextual information like time, location etc. However, there are lots of contextual dimensions that are to be considered in the recommendation systems which may be relevant to predict the rating. In [27], the recommendation system identified the appropriate influence factors among all the important contextual dimensions. The authors have applied fuzzy logic to obtain the current user preference or interest. Feilong Tang et al. [28] proposed a joint aspect-based sentiment topic model, which first extracts the user online reviews and also its associate opinions and then sentiment polarity is calculated for opinions and sentences.

Anh-dung et al [29] improved the efficiency of customer review extraction technique for large datasets. Natural language processing is applied in this system for preprocessing and similar types of domains are considered for analysis. In this system, Laptop and Camera product reviews are considered as the dataset. Many other sentiment analysis methods [30–32] help to identify the opinion of the user reviews, which helps the recommendation system to predict the ratings. However, the method provides the recommendations based on the customer who is purchasing it, instead it is supposed to focus on to whom exactly the product is intended to buy.

Shiliang Sun et al. [33] elaborated NLP techniques to extract the opinions from the customer reviews. To extract the opinion, it is also required to apply preprocessing techniques in NLP like tokenization, POS tagging, Parsing etc. Customer reviews are available in different places with in different formats.

Mondher bouazizi et al. [34] proposed a multi-class sentiment analysis algorithm, which considers all the sentiments that are available in the posted review, instead of only focusing on the single sentiment label. Karthik et al. [35] proposed a recommendation system along with a new Feature and product ranking algorithm (FBPRA). In their algorithm, online customer reviews are evaluated with each features used for sentimental analysis where the sentiment scores are computed for each of the products. Moreover, the products with high score are listed for recommendation to the online users. Based on the outcomes of sentiment analysis based on customer reviews, product recommendations are done. The products that received good feedbacks get higher chances of recommendation to the users. Feature extraction and clustering of the text or customer reviews play an important role in improving the overall performance of the recommendation system. Laith Mohammad Abualigah et al. [36] proposed a hybrid Krill Herd algorithm to effectively cluster the text documents. In [37], the authors further extended the Krill Herd algorithm with a combination of objective functions for clustering text documents.

Laith Mohammad Abualigah et al. [38] proposed a Swarm optimization (PSO) based approach to effectively extract the features from text, which in turn improved the clustering accuracy for text documents and it also reduced the computation time. Kunpeng

Zhang et al. [39] proposed a product ranking model, in which weights are applied for the influence factors and ranking score is evaluated for providing recommendations. Factors like Quality of the product, Duration of the posted review are all considered for computing the overall rating score. Recommendations can be applied to all the application areas. Shahab et al. [40] proposed a technique for recommendation of books by reading the entire user reviews, extracting the product features and categorizing the books. Later on, by applying the feature category, they recommended the user preferred/ interested books.

The recommendation system for E-learning proposed by Sankar et al. [41] helps to recommend the suitable and more relevant contents for e-learning by applying the frequent item patterns and Fuzzy rules that are constructed carefully to accommodate rule mining for all types of learners. This system avoids the chances of exclusion of few item sets considered for recommendation. Many other approaches that can be used for enhancing the accuracy of decision making and recommendations are the Intelligent Fuzzy Rule-Based Recommendation approach for Dynamic User Interests by Sankar et al. [42], Fuzzy Family Tree Similarity Based Effective Recommendation System for e-learning by Sankar et al. [43], a survey of intelligent feature selection and classification techniques by Ganapathy et al. [44], the intelligent risk prediction system using fuzzy temporal rules by Kanimozhi et al. [45], the system developed by Sai Ramesh et al. [46] that focuses on the prediction of user interests for providing relevant information through relevance feedback and re-ranking, the Ontology and Context based Recommendation System using Neuro-Fuzzy Classification developed by Razia Sulthana and Subburaj Ramasamy [47] and the empirical method used for the examination of the influence in biased system personalized product recommendations by Bo et al. [48].

Deborah et al. [49] proposed a logic-based approach for ontology alignment. This approach is useful in the comparison of ontology structures by matching the ontology constructed from the user query and the ontology available in the ontological database. In spite of the presence of many works in the development of recommendation systems, the fuzzy rule based recommendation system proposed in this paper is more efficient with respect to increase in decision accuracy and reduction in time through effective pre-processing, morphological, syntax and semantic analysis, better knowledge representation using ontology and decision making through fuzzy rule based approach, ontology alignment and the handling of incomplete information more effectively.

3. Preliminaries

This section provides the motivation to develop a new recommendation system in the area of e-commerce.

3.1. Scope of improvement in the existing recommendation systems

Most of the existing recommendation systems provide recommendations based on the customer who is going to purchase. Instead, the recommendation system has to focus on the real end user for whom exactly the product is intended to be purchased. Computing rating score and applying it uniformly to all the customers would affect the predicted recommendation list. Ideally, each user age group has different levels of interest on the same product. For example, “Barbie doll” is highly interested by girl child when compared to a boy child of the same age. Existing recommendation systems are focusing more on the similar products that were already purchased by the customer. This would result in the recommendation of the same kind of products (similar) repeatedly.

Existing recommendation systems require improvements in below aspect

- To focus on the real end user: for whom exactly the product is intended to be purchased
- To compute the product rating based on each user group: consider different level of interest
- To consider new products in the recommendation: which are new and interesting to the customer
- To provide variation in the recommendation list: To meet the users varying expectation

3.2. Motivation to develop the proposed new approach

The following factors are considered in the development of the proposed new approach, which overcomes the existing problems and also improves the performance of the recommendation system further:

- The target end user is considered and products are recommended based on the product rating score of that associated real target end user
- Rating score is calculated for the different user category group by referring the computing review to meet different level of interest
- In addition to similar products, considerable percentage of new products are also considered in the recommendation list
- Keeping track of the recommendation history to propose a new set of recommended products for the customer
- Demographic information such as end user age group and delivery location are applied to further improve prediction and get more relevant products
- Intelligent approach with ontology-based knowledge representation and reasoning is necessary for enhancing the decision accuracy in the development of product recommendation systems

4. System architecture

The proposed system contains seven major modules, namely customer (online shopping user), User interface module, Decision manager, Fuzzy Recommendation System, Ontological database, Rule base and Fuzzy rule manager.

The overall architecture of the proposed recommendation system is shown in Fig. 1. The customer sends a request to the recommendation system for recommending a product. The relevant review comments for the requested products may or may not be available in the ontological database. If it is available, then it will be considered for decision-making and will be forwarded to the recommendation system.

The recommendation system consists of three major sub modules such as data preprocessing, sentiment score computation and fuzzy rule-based classifier. The steps used in the decision-making process are as follows:

- Step 1:** Compute the product review score for the product with associated target user age group from preprocessed customer reviews and store the score in the database.
- Step 2:** Obtain the user known products from the purchase history, wish list and search list.
- Step 3:** Extract the high rating products and compute the similarity score.
- Step 4:** Use demographic information age group, delivery location and ontology alignment to improve the recommendation list.

Step 5: Using Fuzzy logic, Generate Highly recommended list & likely recommendation list and display the final recommendation list.

The computation of the product score (Step 1) plays an important role in this proposed recommendation algorithm and it involves more calculations when compared to other steps of this algorithm. First, all the reviews captured from the end users are preprocessed. The data-preprocessing model uses the proposed data pre-processing algorithm called target user-based data pre-processing algorithm for performing the processes of tokenization, parsing and parts of speech (POS) tagging. The proposed algorithm uses NLP for performing the basic pre-processing and it also identifies the relevant and target customer or user. Second, the sentiment score is computed for the pre-processed reviews.

Sentiment analysis is applied to compute the sentimental score of each review comments. This computed sentiment score along with the average customer review score, number of reviews and associated target user category are used to calculate the overall product rating score with respect to each end user group. This product rating score can be used directly to compute recommendation list but, it is required to perform further filtering for customized recommendation.

Step 2 determines the list of preferences by referring the past product history, wish list and the search history. Usually, all the e-commerce sites maintain these details. This helps to predict the user interest and also to understand the user's purchase behavior. After getting the reference list, Step 3 is used to compare the user preference list and the list of products having highly rated product scores. Similarity score is computed to clearly identify two different categories namely the products likely to be known to the end user and the products that are not known to the users. It is not a good option to completely eliminate the products that are not known to users. A good recommendation approach should also include the products that are not known to user. Demographic information filtering is applied for all the products that are not known to the user. It filters the products based on similar age group and delivery location.

Finally, fuzzy logic is applied in Step 4 to compute the final recommendation list. Based on the score, the fuzzy rules are generated according to the sentiment score and are stored in the rule base. The rule manager is used to manage the rules by extracting the necessary rules and sending it to the decision manager for making effective decisions over the review comments. Finally, the recommendation system recommends the highly relevant product to the customer with less time through a user interface module for the customer.

5. Proposed work

This section describes in detail about the methodology and the various steps involved in the proposed recommendation algorithm. Section 5.1 explains the pre-processing technique applied on customer reviews and computation of product rating score with associated target user category. Section 5.2 explains computation of similarity score and group the similar and new products in the recommendation list. Section 5.3 describes the demographic filtering approach to get the highly relevant products. Section 5.4 explains more about how fuzzy logic and fuzzy rules are applied to make the final recommendation list. All the customer reviews are preprocessed and product rating score is calculated. This step is a continuous activity and is applied for all the new customer reviews and feedbacks. Recommendation list is computed on the fly based on the user current interest.

Fig. 2 depicts the detailed structure of the proposed fuzzy logic and sentiment analysis based recommendation model.

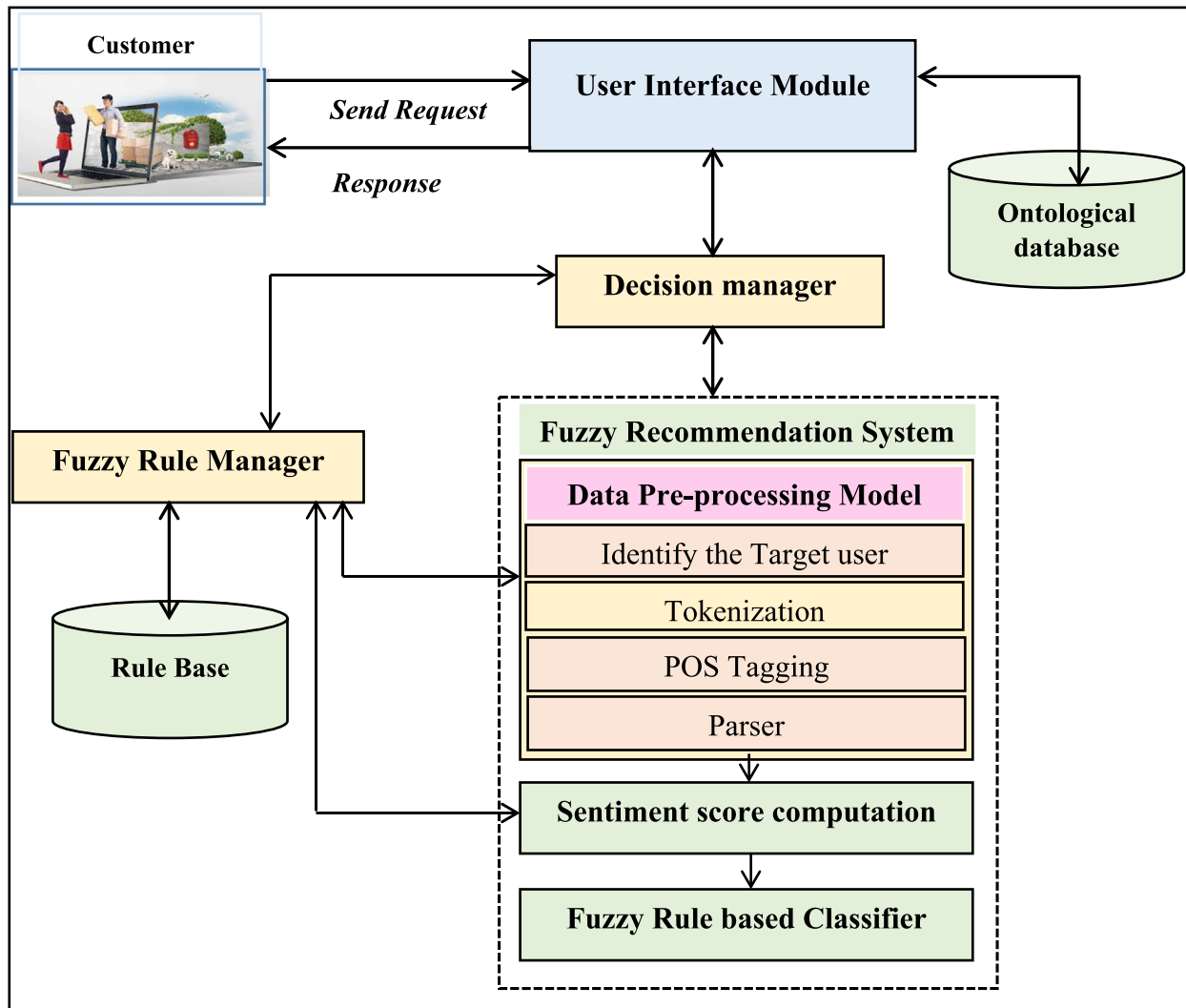


Fig. 1. System architecture.

5.1. Obtain product rating score

This section describes in detail about the data preprocessing, sentiment score computation and product rating processes used in the proposed model.

5.1.1. Data pre-processing

In this system, the data pre-processing is performed by incorporating the activities such as tokenization, POS tagging and parsing. First, the tokenization and morphological analysis phase splits the review comments into tokens, removes all stop words and useless information. Next in the process of POS tagging, the subjective terms are removed and adjectives are considered for further process. Finally, the parsing process is applied for constructing a tree for each review comment. It also implicitly identifies the relationships among the words in the review comments.

A. Tokenization

The contents are divided into tokens or also called as terms. Sequence of words is grouped together such that they give useful semantic units for further processing. In this work, we have provided the user reviews namely, "Awesome product", "Fine look", "My son loved very much" and "Worth for money" to perform tokenization and morphological analysis. Therefore, the

Tokenization step of this algorithm splits a given review into tokens namely, {'awesome','product','fine','look','my','son','loved','very','much'}. Moreover, the stop words are removed from the user review in the tokenization and morphological analysis process of this work.

B. POS Tagging

POS provides the information on how words are used in a sentence. Parts of speech like nouns, pronouns, adjective, and verbs are tagged on tokens in this process. Adjectives play an important role to know the opinion expressed in the comment. POS tagging simply labels the word with its appropriate POS tagging, which helps to know the sentiment/opinion available in the sentence/ comments in further processing. In this work, we have provided user reviews namely "Awesome product", "Fine look", "My son loved very much" and "Worth for money" for part of speech tagging. Each of the tokens are tagged with their part of speech, which helps later to identify the appropriate words for opinion mining. Table 1 shows the POS tags captured for each tokens in the review.

C. Parsing

A Parser is simply a program that provides the grammatical structure for the sentence. The parser model used in this work is grouping together words based on the subject or object. In this work, the tokens with POS tagging are provided as input to the parser and it constructed the parse tree shown below:

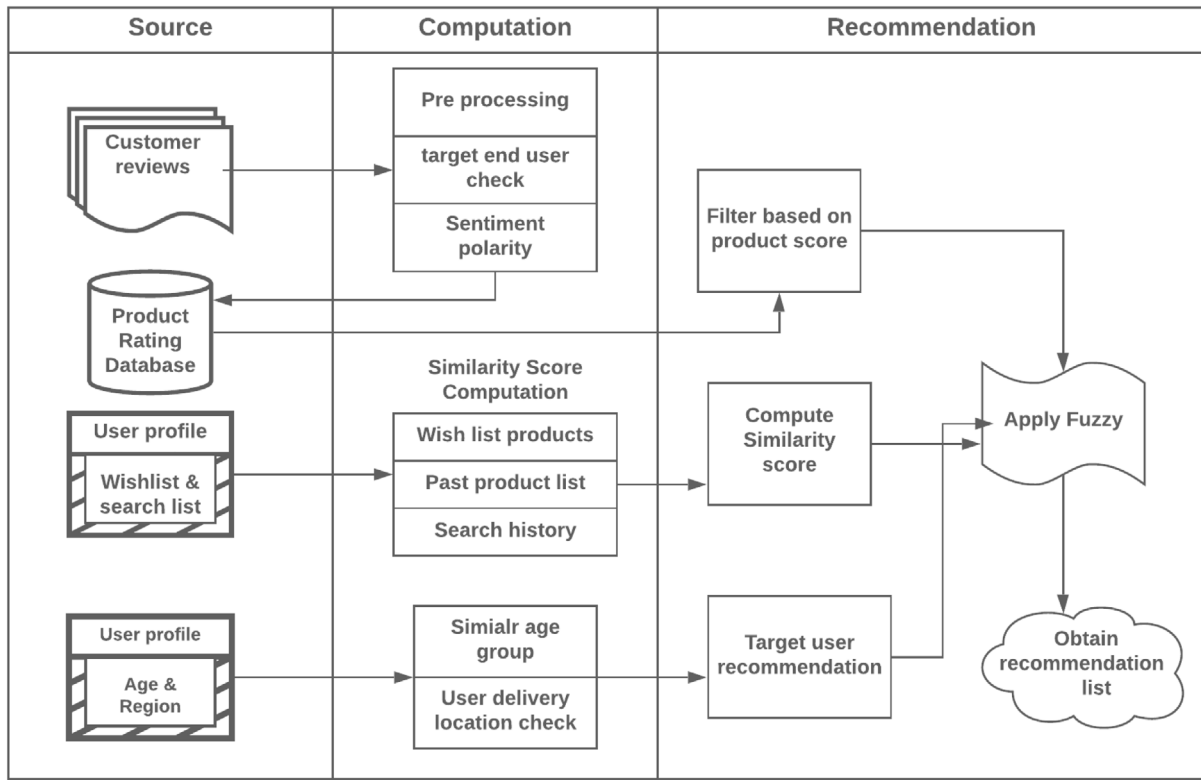


Fig. 2. Proposed recommendation model.

Table 1
POS tagging.

Tokens	POS Tagging	Description
Awesome	JJ	Adjective
Product	NN	Noun
Fine	JJ	Adjective

(ROOT (FRAG (NP (NP (JJ Awesome) (NN product) (JJ Fine) (NN look)) (SBAR (S (NP (PRP\$ My) (NN son)) (VP (VBD loved) (NP (NP (ADJP (RB very) (JJ much)) (NNP Worth)) (PP (IN for) (NP (NN money)))))))))).

The steps of the proposed Target User based Data preprocessing Algorithm (TUDPA) are as follows:

The proposed TUDPA is applied for performing pre-processing over the review comments that are considered from the Amazon dataset. Here, we have used NLP Toolkit for performing the basic pre-processing activities such as tokenization, POS tagging and parsing. The proposed pre-processing algorithm identifies the target user by scanning the review comments and it checks whether the target user details are present in review comments. The major contribution of this proposed data pre-processing algorithm is that it identifies the target user and also performs the pre-processing.

5.1.2. Sentiment score computation

In this section, the product rating score is computed for each preprocessed review by applying the Target User based product rating computation algorithm. First, the Sentiment score (SS) is calculated by considering sentiment polarity for each customer review. Second, Product Rating Score is calculated by summing all the sentiment scores for the associated target user category. In order to improve the sentiment score computation accuracy, we have used five sentiment polarity levels, such as positive, very positive, neutral, negative and very negative.

The working process of the sentiment analysis algorithm proposed in this research work is explained diagrammatically in Fig. 3. The steps of the proposed Target user based sentiment score computation algorithm (TUSAA) are as follows:

Similarly, the review sentiment scores pertaining to associated target users are computed for various products and are stored in the ontological table.

5.1.3. Product rating

Computed sentiment score for each user category is used to compute the product rating score. First average sentiment score $SC_{p,u}$ is computed

$$SC_{p,u} = \sum_{i=1}^{n_{p,u}} (SSi_{p,u}) \quad (1)$$

where $n_{p,u}$ is the number of review for respective user category

Product rating $PR_{p,u}$ for each user category is then calculated with below equation

$$PR_{p,u} = \frac{OverallRating_p * NoOfReview_p}{SC_{p,u}} \quad (2)$$

where $OverallRating_p$ is the overall review rating score of the product 'p' and $NoOfReview_p$ is the number of review of the product 'P'.

Table 2 shows the computation of product rating score $PR_{p,u}$ for one of the product with respect to each target user category, where overall rating of the product is 5 and total number of review is 12.

5.2. Handling of similar and new products

To get the user preference and interest, the products purchased history is considered in this work. Based on product category and the item purchased, similarity score is calculated

Algorithm : Target User based Data Pre-processing Algorithm (TUDPA)	
Input	user review $\{rc_1, rc_2, rc_3, \dots, rc_n\}$
Output	Preprocessed review comments $\{prc_1, prc_2, \dots, prc_n\}$
Step 1: Identify the target user category from the regular expression pattern	
Step 2: Split into tokens using NLP tokenization and remove stop words	
Step 3: Apply POS tagging for each token	
Step 4: Remove unwanted / less information tokens	
Step 5: Construct semantic parser tree using NLP parser	
Step 6: Store the preprocessed review contents	
Step 7: The target user details are stored and can be considered for further processing	

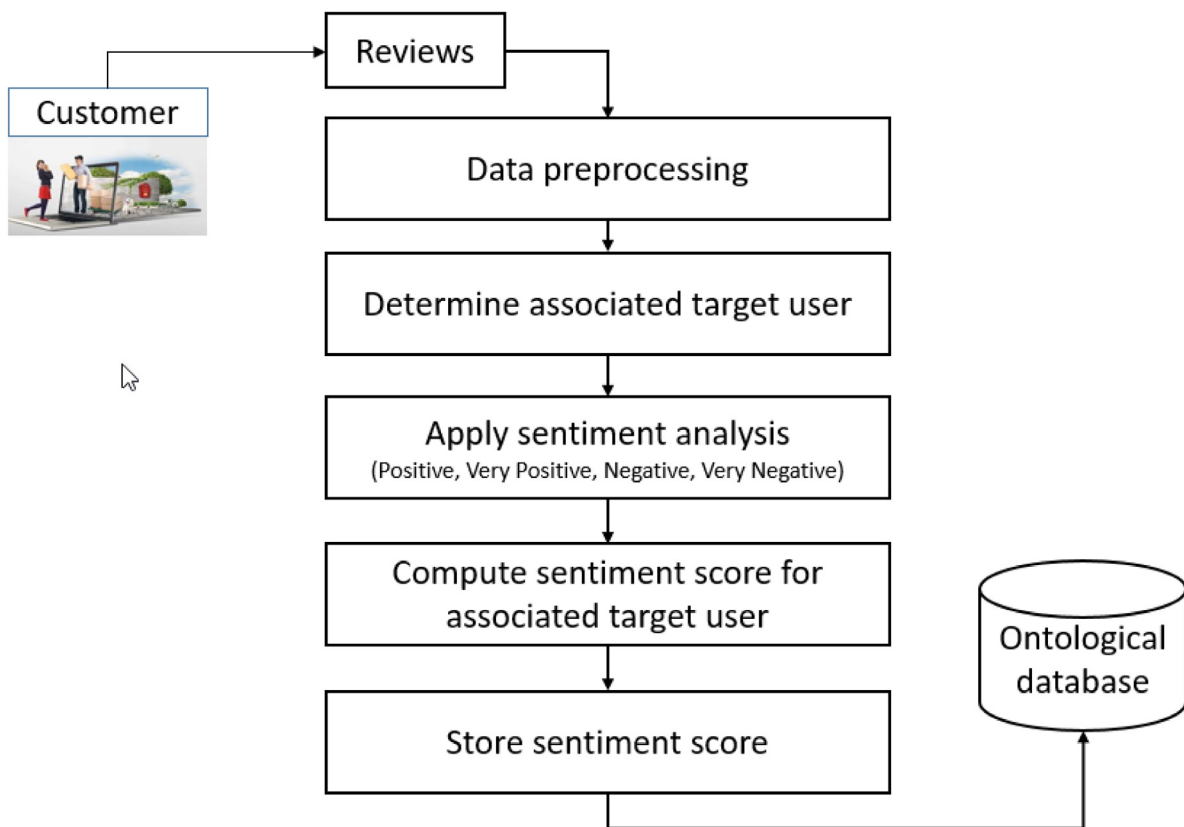


Fig. 3. Sentiment analysis and score computation framework.

Table 2
Product rating score computation.

Product	Computation	Son	Brother	Daughter	Sister
$product_i$	No of reviews	4	4	4	1
[Horse drawn wedding coach-30495]	Average Sentiment Score	0.375	3.125	1.75	5
	Product rating score $PR_{p_i, u}$	0.025	0.208333333	0.116666667	0.083333

for the recommended list of products. More the similarity, more likely that the user is interested to buy the product. Similarity score is computed for the top product rating products against the

past reference products. All the new products are less relevant to the customer when compared to the existing purchase behavior. Still it is believed that the inclusion of those new products could improve the diversity of the recommendation list. The products with greater similarity score are eliminated so that more number of similar products within the recommendation list are avoided. Second, the products that are similar to previous purchase behavior are given more priority when compared to the other products.

Fig. 4 shows the grouping of similar products and new products based on similarity score. The similarity score between two products p_i and p_j is computed by using the cosine similarity

Algorithm: Target User based Sentiment Score Computation Algorithm (TUSSCA)	
Input	Preprocessed review comments {prc ₁ , prc ₂ ... prc _n }
Output	Sentiment scores for each customer review for respective product
Step 1: Read each preprocessed review comments from TUDPA PRC {} = {prc ₁ , prc ₂ , prc ₃ ...prc _n }	
Step 2: Retrieve associated target user category <i>uc</i> from ontological database for each comment	
Step 3: Apply Sentiment for each review comments through NLP	
Step 4: Compute review sentiment score for each review	
Set SC to 1 if positive or set to 2 if very positive sentiment polarity	
Set SC to -1 if negative or set it to -2 if very negative sentiment polarity	
Set SC to 0 if neutral	
Step 5: Compute overall sentiment score for each associated target end user category for all the customer reviews	
$SC_{p,u} = \sum_{i=1}^n ss_i$	
Step 6: Store the computed product rating score with review comment, product, and target user category into ontological database.	

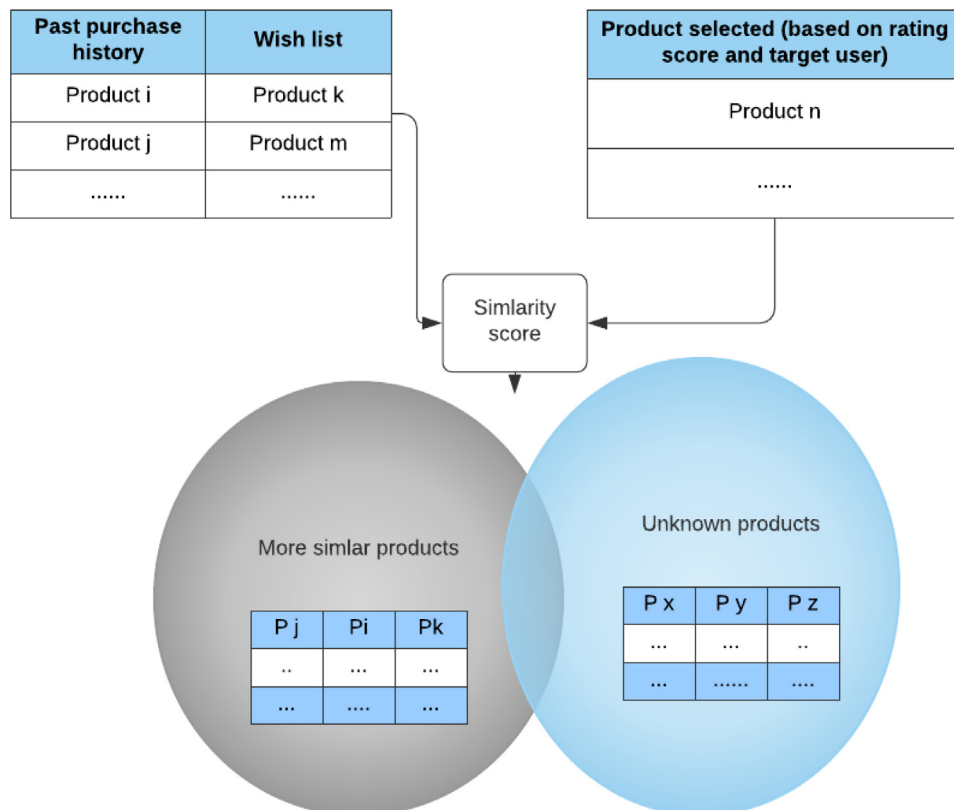


Fig. 4. Computing similarity score and grouping.

computation formula given in Eq. (3).

$$\text{Cosine}(p_i, p_j) = \frac{p_i \cdot p_j^T}{\|p_i\| * \|p_j\|} \quad (3)$$

In order to provide the variation in the recommendation of the product, more similar kind of products are grouped and only few highly rated new products are considered in this work. In the proposed approach, 70:30 ratios are considered for providing the recommendations. Therefore, 70% of the recommended products are relevant to the user browse history and past purchased history, whereas another 30% of the recommendation products are new to the customer.

5.3. Applying demographic information

The computed recommendation list based on the similarity score contains both known and new products to the customer. Presence of new products in the list would affect the relevancy factor and hence it will reduce the confidence level of the recommendation system. On other hand, preferring only the known products would not meet the customer's new interest. Demographic information filtering method is applied to overcome this problem in the proposed system. It is based on customer age group, associated products are filtered from the new products. This would reduce the impact of the presence of unknown products in the recommendation list.

5.4. Fuzzy logic

This section explains in detail about the fuzzy logic, which is applied in this work, in the process of generating fuzzy rules that are used in the computation of sentiment score and analysis. In this product recommendation system, there are possibilities that few average level products may also be interested to the users. Instead of applying the crisp logic to provide the decisions such as recommended and not recommended, the output can be provided with varying levels of decisions by applying fuzzy logic. Therefore, this system uses five linguistic variables namely highly recommended, Medium Highly Recommended, Recommended, Likely to be recommended and not recommended. This will improve the accuracy and efficiency of recommendation system.

Fuzzy logic is mainly relied on the working principle of membership function. It defines the input and output datasets. It is important to use the appropriate membership function type and parameter for membership functions that suite for the dataset. It helps to quantify linguistic term and denote a fuzzy set realistically. There are various types of membership functions available: (i) Triangular: The triangular membership function work on three scalar parameters 'a' and 'c' locate the base of the triangle and the parameter b locates the peak by (ii) trapezoidal: The Trapezoidal membership function operate on four scalar parameters 'a', 'b', 'c' and 'd'. The parameters are defined by a lower limit a, an upper limit d, a lower support limit b, and an upper support limit c, where $a < b < c < d$, iii) sigmoidal, (iv) Gaussian: Membership function is usually represented as $\text{Gaussian}(x;c,s)$ where c, s represents the mean and standard deviation. Here, c represents center, s represents width and m represents fuzzification factor.

In this proposed new recommendation system, four influence factors such as Sentiment Score, Overall product score, Target User Recommendation Score and Recommendation Hit rate have been considered for recommending the suitable products. Moreover, the output is the recommended level and the output membership functions are designed based upon the computed score. Also considering the faster computation and simple and efficient factors, it is advisable to use Triangular Membership

Table 3
Fuzzy internal factors.

Factor	Base 1	Base 2	Outcome
Sentiment score	0	3	very negative
	2.5	5.5	negative
	5	8	positive
	7	10	very positive
Product rating score	0	2.5	poor
	2.3	3	average
	2.8	4	good
	3.8	5	excellent
Similarity score	0	3	poor
	2.5	5.5	average
	5	8	good
	7	10	excellent

function considered for all the fuzzy variables are constructed. Here, trapezoidal fuzzy membership is used in this work.

The trapezoidal membership function consists of four parameters {a, b, c, d} and it shows in Eq. (4).

$$\text{trapezoid}(x; a, b, c, d) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a \leq x \leq b \\ 1, & b \leq x \leq c \\ \frac{d-x}{d-c}, & c \leq x \leq d \\ 0, & d \leq x \end{cases} \quad (4)$$

The same fuzzy membership function is also can be defined as given in Eq. (5).

$$\text{trapezoid}(x; a, b, c, d) = \max\left(\min\left(\frac{x-a}{b-a}, 1, \frac{d-x}{d-c}\right), 0\right) \quad (5)$$

The fuzzy membership function parameters {a, b, c, d} with a less than b, b less than or equal to c, c is less than d that determines the x coordinate values of the four possible sides of the specific trapezoidal fuzzy membership function.

Moreover, the fuzzy logic approach is applied to convert product sentiment score to the recommendation level by applying Fuzzy rules for the respective target user. Fuzzy rules are written in such way to categorize the recommendation level based on the review sentiment score.

5.4.1. Fuzzy interval

A fuzzy interval is a fuzzy set in the real line whose level-cuts are intervals. Table 3 represents the Fuzzy interval applied for sentiment score, overall product rating and similarity score respectively.

5.4.2. Fuzzy rule and role of fuzzy rules

Fuzzy rules are framed by considering the overall product rating, calculated target product score, similarity with history and purchased period.

The above-mentioned fuzzy rules are applied for making effective decisions over the review comments of every online user's and their comments. These kind of various fuzzy rules are stored in the ontological database. When the situation has arisen then these rules will be taken consider and applied. If a customer is looking to purchase the product to his/her son, then the proposed recommendation system will recommend the products by referring the recommended level. All the highly recommended products are provided to the user, followed by recommended products.

5.4.3. Fuzzy recommendation system

This section explains the overall process of the proposed Target user-based product rating recommendation algorithm used in recommending the relevant products for appropriate target

Algorithm: Target User based Data Pre-processing Algorithm (TUDPA)	
Input	Let $X = \{x_1, x_2, x_3, \dots, x_n\}$ products that available in purchased history and browse history Let $y = \{p_1, p_2, p_3, \dots, p_n\}$ recommendation products
Output	Product recommendation list (PRL) for associated end user category
Step 1: Fetch High product rating score products Step 2: Develop (x,y) associated map Step 3: For each map entry do Calculate cosine similarity value $\text{Cosine}(p_i, p_j) = \frac{p_i \cdot p_j^T}{\ p_i\ \cdot \ p_j\ }$ Calculate the weightage of the dissimilar relevant product $(W_{us})^n = \sum_{k=0}^n PC_{ij} * W_{fij}$ where, $\{C_1, C_2, \dots, C_n\}$ is the category of the product Reduce Key and store the computed value in map End Step 4: Cluster the products based on computed score Step 5: Remove the product from the recommendation list if it is already purchased Step 6: Store the computed list (PRL) in Ontological database	

user categories. First, it reads all the review comments for each product. Second, it pre-processes all the review comments for each product and the associated target user by applying the Target User-based Data Preprocessing Algorithm (TUDPA). Third, it applies the sentiment analysis and computes the product rating score for each product using the proposed Target user-based sentiment score computation algorithm (TUSSCA). Fourth, it identifies the recommendation level by applying Fuzzy rules. Finally, recommend the product based on recommendation level.

Target User based Recommendation Algorithm (TUSSCA)

Input Online review comments and target user

Output : Recommended product for the requested target user

Step 1: Read all the reviews for each product $\{P_1, P_2, \dots, P_n\}$

Step 2: Apply preprocess for all the reviews and identify associated target users for each review using Target user-based data preprocessing algorithm (TUDPA)

Step 3: Identify sentiment polarity for each review

Step 4: Compute overall product rating score for each associated target user category by using Target user-based sentiment score computation algorithm (TUSSCA)

Step 5: Compute similarity score and cluster similar and unknown products

Step 6: Apply demographic filter based on age group to filter unknown products

Step 7: Apply fuzzy rules and compute the recommended level

Step 8: Perform ontology alignment to analyze the recommendation and the level

Step 8: Recommend the highly recommended product to the user for the given target user category

This proposed algorithm recommends products by performing sentiment analysis for customer reviews with associated target user. The major contribution of this proposed fuzzy logic-based recommendation algorithm is that it identifies the products that are suitable for the target user where the customer really wants to purchase. Here, fuzzy rules are applied for making effective decisions over the sentimental score of the review comments that are pre-processed. The recommendations are finally crosschecked, evaluated and improved using ontology alignment. This proposed fuzzy recommendation system recommends the product according to the sentimental score as “Not recommended”, “Likely to recommend”, “Recommended” and “High recommended products”.

6. Results and discussion

In this work, six different data sets are considered for experiment. Dataset 1 includes Books amazon customer review dataset for evaluating the proposed product recommendation algorithm with other recommendation systems. Dataset contains different product with more than 51, 00,000 reviews. Dataset 2 includes **Movies and TV with more than 8,700,000** reviews. Toys and Games amazon customer reviews are available in Dataset 3 with more than 8,200,000 reviews. Dataset 4 is constructed by combining both **Books and Toys** and **Games** amazon dataset. Similarly Dataset 5 is constructed by combining both **Toys and Games** and **Books** amazon dataset. Dataset contains product name, product description, product cost, product overall review rating, similar products and customer reviews. Dataset 6 contains **Movies and TV** along with Toys and Games and Books dataset [50]. Combining

Fuzzy Rules	
RULE 1	IF Sentiment_Score= Very positive AND Overall_Product_Rating= Excellent AND Past_Purchase_Similarity= Excellent AND Target_User_Score= Excellent AND Purchased_Month is April or May THEN Recommendation_Level= Highly Recommended
RULE 2	IF Sentiment_Score= Very positive AND Overall_Product_Rating= Excellent AND Past_Purchase_Similarity= Good AND Target_User_Score= Good AND Purchased_Month is April or May THEN Recommendation_Level= Highly Recommended
RULE 3	IF Sentiment_Score= Positive AND Overall_Product_Rating= Good AND Past_Purchase_Similarity= Good AND Target_User_Score= Good AND Purchased_Month is April or May THEN Recommendation_Level= Highly Recommended
RULE 4	IF Sentiment_Score= Positive AND Overall_Product_Rating= Average AND Past_Purchase_Similarity= Excellent AND Target_User_Score= Excellent AND Purchased_Month is April or May THEN Recommendation_Level= Highly Recommended
RULE 5	IF Sentiment_Score= Positive AND Overall_Product_Rating= Average AND Past_Purchase_Similarity= Good AND Target_User_Score= Good AND Purchased_Month is April or May THEN Recommendation_Level= Recommended
RULE 6	IF Sentiment_Score= Positive AND Overall_Product_Rating= Average AND Past_Purchase_Similarity= Average AND Target_User_Score= Excellent AND Purchased_Month is April or May THEN Recommendation_Level= Recommended
RULE 7	IF Sentiment_Score= Positive AND Overall_Product_Rating= Excellent AND Past_Purchase_Similarity= Good AND Target_User_Score= Good AND Purchased_Month is NOT April or May THEN Recommendation_Level= Recommended
RULE 8	IF Sentiment_Score= Negative AND Overall_Product_Rating= Average AND Past_Purchase_Similarity= Average AND Target_User_Score= Good AND Purchased_Month is April or May THEN Recommendation_Level= Likely to recommend
RULE 9	IF Sentiment_Score= Positive AND Overall_Product_Rating= Average AND Past_Purchase_Similarity= Average AND Target_User_Score= Average AND Purchased_Month is April or May THEN Recommendation_Level= Recommend
RULE 10	IF Sentiment_Score= Very Negative AND Overall_Product_Rating= Average AND Past_Purchase_Similarity= Average AND Target_User_Score= Poor AND Purchased_Month is Not April or May THEN Recommendation_Level= Not recommend
RULE 11	IF Sentiment_Score= Very Negative AND Overall_Product_Rating= Poor AND Past_Purchase_Similarity= Average AND Target_User_Score= Poor AND Purchased_Month is April or May THEN Recommendation_Level= Not recommend
RULE 12	IF Sentiment_Score= Positive AND Overall_Product_Rating= Excellent AND Past_Purchase_Similarity= Good AND Target_User_Score= Good AND Purchased_Month is NOT April or May THEN Recommendation_Level= Recommended

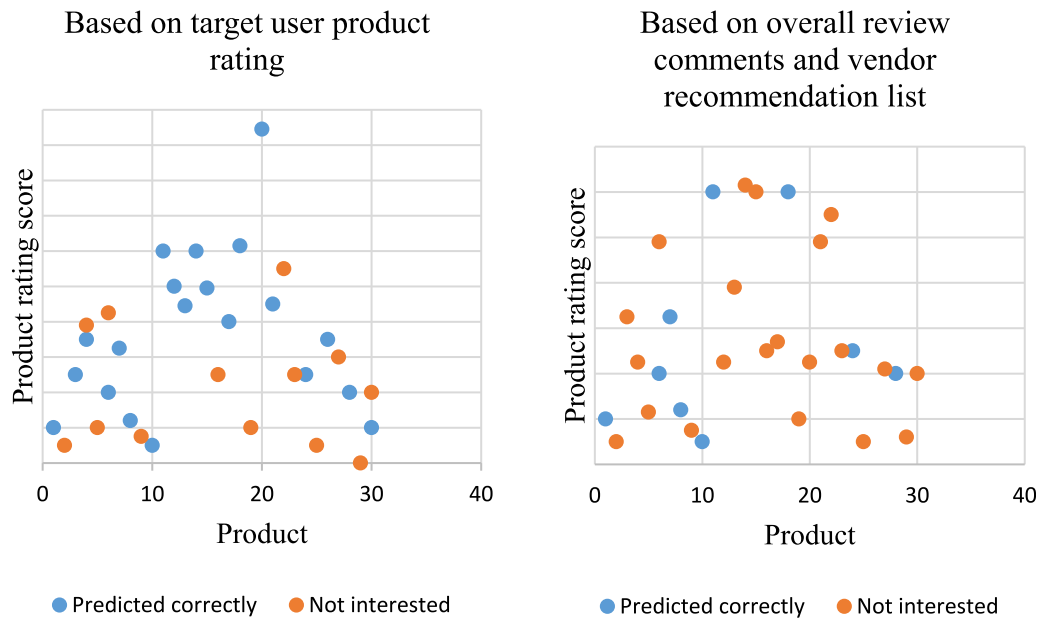


Fig. 5. Evaluating product rating score factor.

Table 4

Dataset details.

Dataset	Category	Number of reviews	Dataset details
DAS1	Books	51,300,000	Review, Overall rating, reviewer id, reviewer name, Product id, related products, categories, Similar products
DAS2	Movies and TV	8,700,000	Review, Overall rating, reviewer id, reviewer name, Product id, related products, categories, Similar products
DAS3	Toys and Games	8,200,000	Review, Overall rating, reviewer id, reviewer name, Product id, related products, categories, Similar products
DAS4	Books + Movies and TV	59,700,000	Review, Overall rating, reviewer id, reviewer name, Product id, related products, categories, Similar products
DAS5	Books + Toys and Games	58,150,000	Review, Overall rating, reviewer id, reviewer name, Product id, related products, categories, Similar products
DAS6	Movies and TV + Books + Toys and Games	68,400,000	Review, Overall rating, reviewer id, reviewer name, Product id, related products, categories, Similar products

the different category dataset helps to evaluate the recommendation algorithm with various types of products. Table 4 shows dataset details. Evaluation results are presented in detail in this section.

6.1. Evaluation method

The test data is split into two halves. One-half is used to refer the history data of the user, and is called as past reference data. Whereas another half of the data used to evaluate the outcome of recommendation algorithm, is called as predictive data. Based on the previous purchase history and product score, this algorithm computes the similar and dissimilar recommended products. These products are checked against the predictive data set whether the recommended product is present in the predictive data set. The presence of recommendation product in the predictive data set provides the confidence and coverage of similar products in the recommendation list. As the proposed algorithm is also covering the dissimilar products for recommendation, the presence of unknown of products in the recommendation list on comparing the predictive data set is also evaluated.

6.2. Preference to target end user

In the proposed algorithm, the product rating score is computed based on associated target user group when the end user

detail is given in the purchase preference. All the products that are not associated to the target users are eliminated in the proposed method and also the products that have poor review comments are not considered. On the other hand, products are predicted based on the overall review score and the products recommended by the vendor. For example: When the customer is interested to purchase the product for his/her son, then proposed method computes the recommendation list based on the target end user "son" with high product rating score. The output of the proposed algorithm provides promising results when compared to other recommendation algorithms that are working based on the overall review comments or by the comparison of the history.

Fig. 5 shows the experiment results and depicts that more than 65% of the products are available in the prediction list of the evaluation method. Whereas the technique with the vendor recommendation list and the overall review score list has less than 40% of the products.

6.3. Considering new products for the recommendation

Predicting the top recommendation list from the huge volume of option is very challenging. Most of the recommendation lists refer to the past history, which results in recommending the similar products already known to the customer. Recommendation of similar kind of products would not be right approach as the user interest may change over time. In addition to similar

products, it is also important to recommend the products that are not known to the customer. This helps when the user interests' are changed or they are willing to purchase some new type of products. In the proposed approach, similarity score is computed for the top product-rating list against the user preference product list (past purchased history and wish list). This similarity score helps to distinguish between products that are known to customer and those products that are not known to the customer. The products that are not known to the customer, but has high good product rating score are potential products that are likely to be purchased by the customer. Recommending these products also gives the diversity in the recommendation list. This helps to give an opportunity to refer the products to which user the user is not exposed. Fig. 6 shows the products that are known to the customer in both the proposed technique and other past history-based recommendation techniques. In the history-based technique, all the predicted products are almost 85% known to the customer.

6.4. Influence of demographic information in the recommendation

Though the similarity score helps to obtain the products with high potential for purchase by the Customer, a challenge still exists in identifying the top recommended product from the big list of identified potential products. In the real scenario, the customer is interested to see only a smaller number of recommendation proposals from the recommendation system. Recommending more than 25 products may reduce the interest on the recommendation system for the customer. Therefore, it is necessary to list out only few products for the recommendation. To address this challenge, demographic information – customer age group are also considered in the proposed technique. This information helps to further filter the products with high rated product score and the similarity score. In the propose technique, products that are not known to the customer are also considered for recommendation which may affect the accuracy of the prediction of products. To address this, demographic information is applied to take only the selected products, which meet the customer's age group or location. From the product sales information available in the database, it is possible to get the age category group, which has purchased the maximum quantity of the respective product. The experimental results depicted in Fig. 7 shows the presence of previously purchased products in the top recommendation list before and after applying the demographic information. By considering the product rating score and similarity score computation, it is found that only 50% of products are potential to purchase by the customer from the top N recommendation list. After applying the demographic information and identifying top N products for the recommendation by considering customer age group, the results show that 70% of the products are potential for purchase by the customer; as those products are available in the evaluation data prediction list.

6.5. Avoid repeated recommendation with help of hit value

In online shopping, the user will not prefer to see the same recommendation list again. Sometimes, the same product is recommended often that does not interest the customer. Hence, repetition of the same product will affect the interest of the recommendation system. To provide the new recommendation list every time, the proposed algorithm uses hit ratio; which helps to fetch next set of recommendation list. This recommendation hit is tracked and maintained in the database for each user in the proposed recommendation system. When the user accesses it, the proposed algorithm prepares the final recommendation list based on the hit ratio value. By providing the new set of recommendations, it is possible to increase the interest on the recommendation system to the customers.

6.6. Performance metrics

This proposed recommendation system has been evaluated based on the standard evaluation metrics such as Precision and Recall defined in the Eqs. (6) and (7).

$$\text{Precision} = \frac{\text{Relevant Recommended Products}}{\text{Total Recommended Products}} \quad (6)$$

$$\text{Recall} = \frac{\text{Relevant Recommended Products}}{\text{Total no. of relevanted products to be Recommended}} \quad (7)$$

In addition to standard metrics, Recommendation specific metrics namely Diversity, Novelty and Serendipity are also considered for evaluation.

Diversity: It helps to measure how diverse is the proposed recommendation algorithm. In case if the recommendation results in high accuracy, then there would be more possibility of proposing the same products repeatedly. This Diversity metrics helps to measure how each proposal differs from earlier. The diversity is calculated by using the formula, which is given in Eq. (8).

$$\text{Diversity} = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \text{Sim}(x_i, y_j) \quad (8)$$

Novelty: Novelty measure helps to identify number of new products that are not known to the in the recommendation list. Higher the novelty value, the proposed recommendation products are less popular.

The novelty value is measured by using the formulae that are given in Eqs. (9) and (10).

$$\text{Novelty} = \sum_{x=1}^n \frac{\log_2 P_x}{\text{no.}} \quad (9)$$

$$\text{NoveltyI}(x) = -\log_2 p(x) \quad (10)$$

where,

$P(x)$ – represents the probability that x is observed

Serendipity: It helps to measure how the proposed products or items give surprise or how it is relevant to the customer interest. Degree of surprise is computed by considering the probability of product x that is recommended to the target end user and with how much probability that same product i is recommended for any other customer. Serendipity is calculated by using the formulae that are given in Eqs. (11) and (12).

$$P_x = \frac{\text{no.} - \text{rank}_x}{\text{no.} - 1} \quad (11)$$

$$\text{Serendipity}_{\text{customer}} = \frac{1}{n} \sum_{x=1}^n \max (P_x (\text{Customer}) - P_x (\text{allcustomers}), 0) * \text{rel}_x(\text{customer}) \quad (12)$$

nDCG: nDCG stands for normalized discounted cumulative gain. This measure helps to verify the correctness of the proposed recommendation product order and it is calculated by using the formula, which is given in Eq. (13).

$$\text{nDCG}(L, k) = \frac{1}{|L|} \sum_{x=1}^{|L|} Z_{kx} \sum_{m=1}^k \frac{2^{R(x,m)} - 1}{\log_2(1 + m)} \quad (13)$$

where, L represents the set of proposed recommendation list and $R(x,m)$ indicates relevance score for respective recommendation.

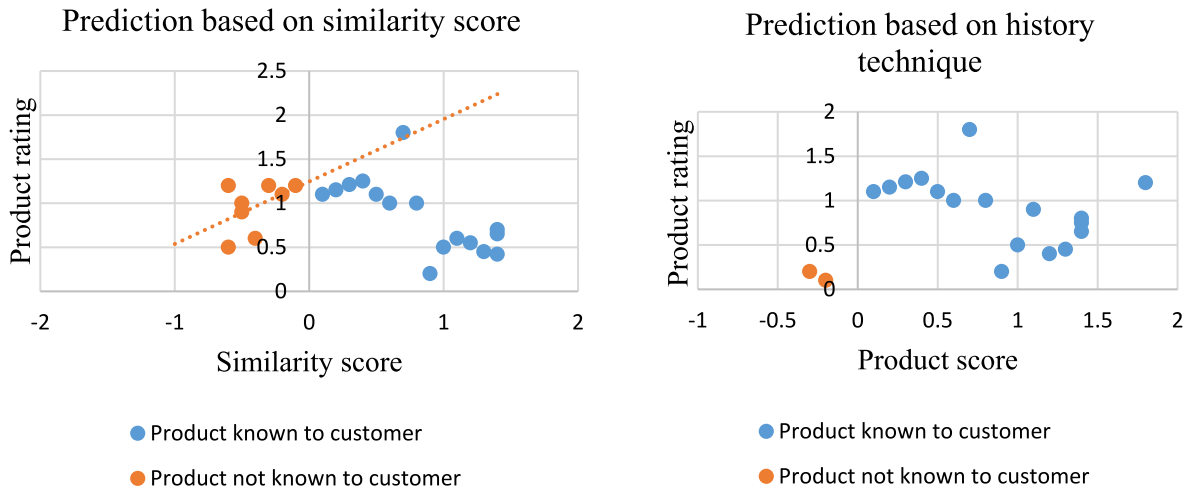


Fig. 6. Evaluating similarity score factor.

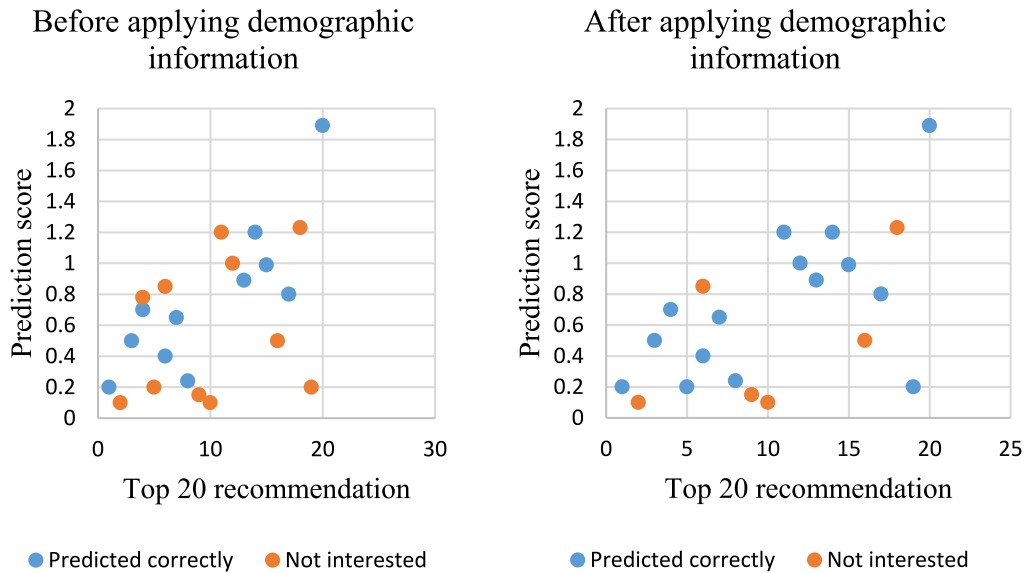


Fig. 7. Evaluating demographic influence factor.

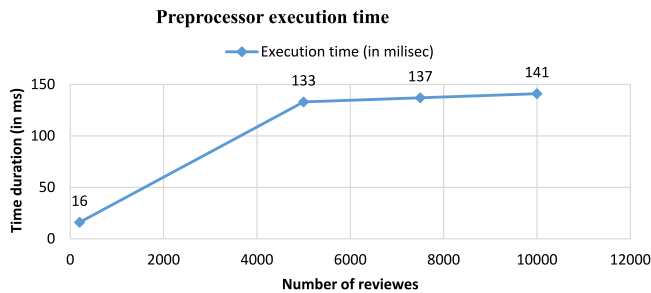


Fig. 8. Preprocessing time analysis.

6.7. Experimental results

Evaluated the execution time of preprocessor with respect to the number of reviews for the proposed target user based data pre-processing algorithm. Here, we have considered the different number of reviews such as 100, 5000, 7500 and 10000 that are randomly considered from the dataset.

From Fig. 8, it can be seen that the execution time of the proposed target user based data pre-processing algorithm takes significant amount of time for the different range of review comments that are considered for conducting six experiments. It is also observed that the execution time gradually increases when the review comments are increased. This is the reason to use target user identification process over the standard preprocessing of NLP.

Figs. 9 and 10 demonstrate the performance comparative analysis between the proposed fuzzy recommendation system and the existing fuzzy recommendation systems that are available in this direction. Precision and recall value are considered for the comparative analysis in this work. Fig. 9 shows the precision value based comparative analysis. Fig. 10 demonstrates the recall value comparative analysis. Experiment conducted with all three datasets DSA1, DSA2 and DSA3 which contain different products.

From comparative analysis on all three metrics, it can be seen that the proposed fuzzy recommendation system performs better than the existing Matrix factorization and content based recommendation system. It also slightly outperform when compare the similar fuzzy based recommendation system (Sankar et al. 2019). This performance improvement is achieved by the incorporation

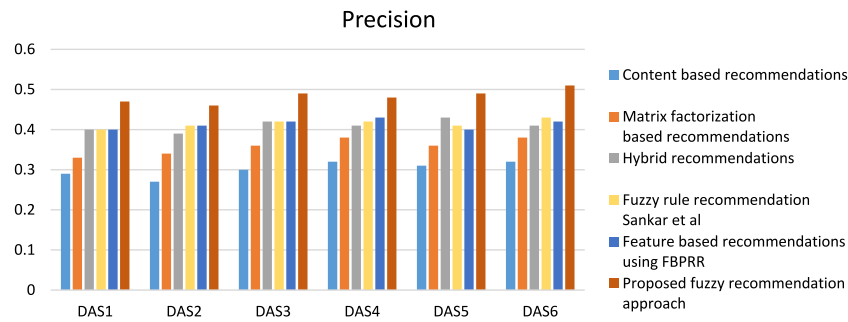


Fig. 9. Precision value based comparative analysis.

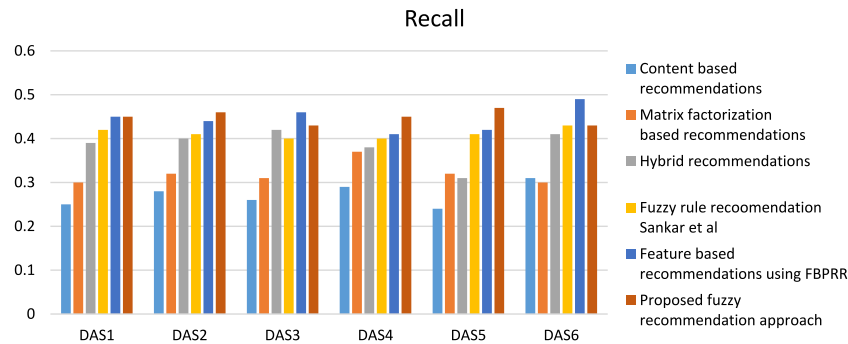


Fig. 10. Recall Value based comparative analysis.

of new fuzzy rules in the proposed fuzzy recommendation system and also using the effective target user based pre-processing algorithm.

Fig. 11 demonstrated that the recommendation specific metrics comparative – Diversity analysis between the proposed fuzzy recommendation system and the existing hybrid recommendation systems that are available in this direction. In the proposed technique, the similarity score plays a significant role in identifying the different products for the recommendation. By considering the unknown products and filtering them based on demographic information helps to identify different unknown products interests the customer to purchase. In addition to similarity score, the proposed technique also considers the recommendation hit ratio to provide different recommended products when the next time customer accesses it. This avoids the recommendation of same product or more similar products again and gives good result for Diversity metrics.

Fig. 12 demonstrates the recommendation specific Novelty metrics based comparative analysis between the proposed fuzzy recommendation system and other Hybrid recommendation system. Novelty focuses on the products that are known to the customer. In the proposed technique, past product history and wish list are considered to identify the similar products and to recommend the products that already known to customer. In the proposed technique, both similarity and Fuzzy rules play significant roles to obtain the top recommendation of products. Though the proposed technique involves both known and unknown products, the fuzzy rules discard all larger and dissimilar score products and they fetch the products that have high potential value of purchase.

Serendipity metrics focus on the new relevant products in the recommendation system. By combining both similarity score and fuzzy rules, the accuracy of the relevant product in the recommendation list is increased in the proposed approach when compared to other hybrid approach.

Fig. 13 demonstrates the results using recommendation specific Serendipity metrics. The possibility of identifying the relevant products outperforms in the proposed technique when the customer specifies the intend target end user.

nDCG metrics focuses on the order of the recommendation list. In the proposed technique, the use of fuzzy inference and the product rating score play very significant role in deciding the order of the recommendation list. Fuzzy logic categorizes the recommendation as highly recommend, recommended and likely to be recommended. All the highly recommended products are ordered first. This techniques improves the recommendation order, as a result, the proposed fuzzy recommendation technique outperforms the other hybrid techniques. This recommendation further improves when the customer specifies the target end users. The proposed method computes the recommendation list based on the product ratings score based on the associated end user.

Fig. 14 demonstrates the recommendation specific nDCG metrics. On considering the unknown and dissimilar products with demographic information, the proposed algorithm based on fuzzy rule is giving optimal results. All the factors considered in the proposed technique have greatly improved the recommended method when compared to the existing recommendation systems. Table 5 shows the consolidated results with all the various set of experiments. The proposed recommendation system shows better result when compared to existing recommendation systems.

7. Conclusion and future work

A new fuzzy logic-based product recommendation system has been proposed and implemented in this paper for recommending the right product to customers in online shopping. The proposed recommendation system involves four important steps including the computation of product rating score based on target user. By considering the known or unknown products using the similarity score, the proposed system applies demographic age category

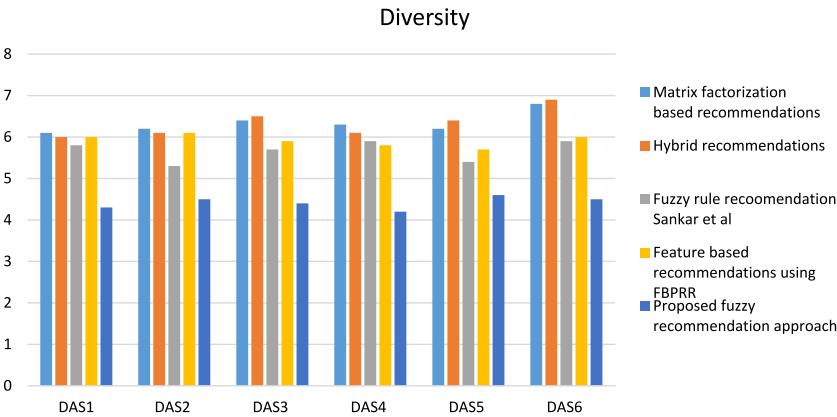


Fig. 11. Diversity based comparative analysis.

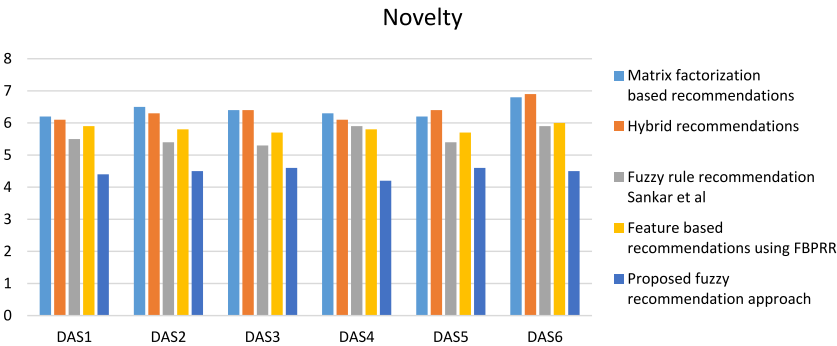


Fig. 12. Novelty based comparative analysis.

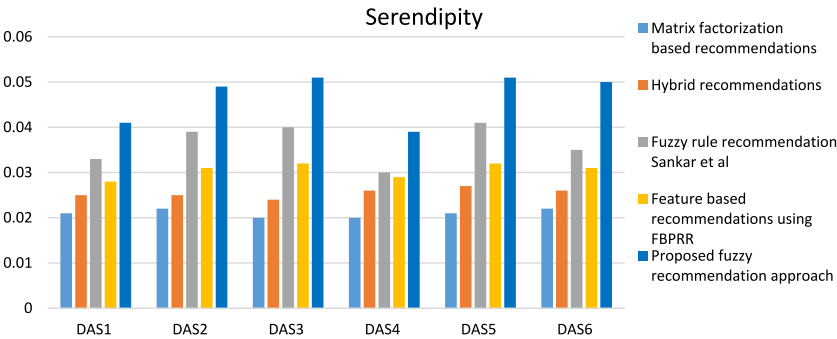


Fig. 13. Serendipity based comparative analysis.

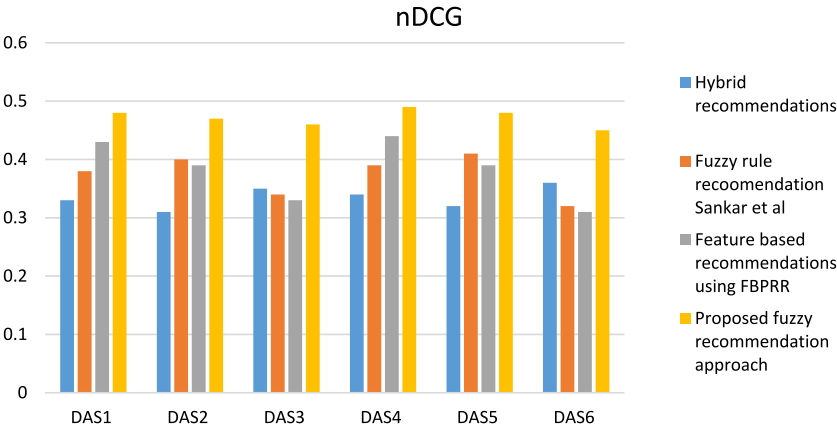


Fig. 14. nDCG based comparative analysis.

Table 5
Comparative analysis.

Dataset-Top N	Method	Precision	Recall	Diversity	Novelty	Serendipity	nDCG
DAS1 -Top10	Matrix factorization based recommendations	0.33	0.30	6.1	6.2	0.021	–
	Hybrid recommendations	0.40	0.39	6.0	6.1	0.025	0.33
	Sankar et al	0.40	0.42	5.8	5.5	0.033	0.38
	FBPRR	0.41	0.45	6.0	5.9	0.028	0.43
	Proposed fuzzy recommendation approach	0.47	0.45	4.3	4.4	0.041	0.48
DAS2 -Top10	Matrix factorization based recommendations	0.34	0.32	6.2	6.5	0.022	–
	Hybrid recommendations	0.39	0.4	6.1	6.3	0.025	0.31
	Sankar et al	0.41	0.41	5.3	5.4	0.039	0.40
	FBPRR	0.41	0.44	6.1	5.8	0.031	0.39
	Proposed fuzzy recommendation approach	0.46	0.46	4.5	4.5	0.049	0.47
DAS3 -Top10	Matrix factorization based recommendation	0.36	0.31	6.4	6.4	0.02	–
	Hybrid recommendations	0.42	0.42	6.5	6.4	0.024	0.35
	Sankar et al	0.42	0.4	5.7	5.3	0.04	0.34
	FBPRR	0.43	0.46	5.9	5.7	0.032	0.33
	Proposed fuzzy recommendation approach	0.49	0.43	4.4	4.6	0.051	0.46
DAS4 -Top10	Matrix factorization based recommendations	0.38	0.37	5.9	6.3	0.02	–
	Hybrid recommendations	0.41	0.38	6.56	6.1	0.026	0.34
	Sankar et al	0.42	0.4	4.9	5.9	0.03	0.39
	FBPRR	0.43	0.41	5.9	5.4	0.029	0.44
	Proposed fuzzy recommendation approach	0.48	0.45	4.8	4.2	0.041	0.49
DAS5- Top10	Matrix factorization based recommendations	0.31	0.32	6.1	6	0.021	–
	Hybrid recommendations	0.43	0.31	6	6.4	0.027	0.32
	Sankar et al	0.41	0.40	4.9	5.4	0.041	0.41
	FBPRR	0.43	0.42	6.1	6.1	0.032	0.39
	Proposed fuzzy recommendation approach	0.49	0.47	4.6	4.6	0.051	0.48
DAS6-Top10	Matrix factorization based recommendations	0.32	0.31	6.2	6.8	0.022	–
	Hybrid Recommendation System	0.41	0.40	5.2	6.9	0.035	0.36
	Sankar et al	0.43	0.42	5.2	5.9	0.031	0.32
	FBPRR	0.42	0.49	5.4	6.0	0.03	0.31
	Proposed fuzzy recommendation approach	0.48	0.46	4.7	4.5	0.05	0.45

filter and finally fuzzy logic-based product recommendation classifier. The proposed technique calculates the product rating score based on the review comments with the associated target user. This has improved the recommendation system in predicting the product when compared to the recommendations provided by the respective vendors. Consideration of the negative sentiment in the customer reviews helps to improve the product rating score.

One of the biggest challenges addressed in the proposed recommendation system is the capturing and consideration of user interests that are changing over time. This challenge is addressed in the proposed technique, by determining both known and also unknown products to the customer by applying the similarity score. The proposed technique includes the products that are not known to the customer, but with a high degree of potential to buy. In other hand, unknown products in the recommendation system should not be a surprise to customer. So, demographic information and purchaser age group are considered in this work to filter the relevant products for the recommendation, which improves the recommendation list. Finally, fuzzy rules are applied to predict the highly relevant recommendation products by performing sentiment analysis and ontology alignment in the decision-making process. The experimental results of the proposed recommendation system have proven that it is better than the existing systems in terms of prediction accuracy of products and target users. Specifically, it works based on target user demographic information (to whom the customer is interested to purchase), customer location and purchase period.

Future works can be done in this work by the introduction of intelligent Neuro-fuzzy approach for making effective decisions and new pre-processing technique.

CRediT authorship contribution statement

R.V. Karthik: Conceptualization, Designed the analysis, Data curation, Contributed data or analysis tools, Formal analysis,

Writing - original draft. **Sannasi Ganapathy:** Contributed data or analysis tools, Formal analysis.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: We (R.V Karthik & Sannasi Ganapathy) are interested to publish this research work in your esteemed journal.

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